
LiveTS: Release-Date-Stratified Retrospective Evaluation of Zero-Shot Time Series Forecasting

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(LiveTS working draft; author list to be finalized)

Abstract

Time series foundation models (TSFMs) are now routinely evaluated in the “zero-shot” regime on benchmarks that predate their pretraining corpora. This creates two failure modes that no amount of statistical care can repair post hoc: (i) *pretraining contamination* — the test data may literally be in the training set — and (ii) *community overfitting* — static test sets are reused across thousands of experiments, turning reported gains into selection effects. Live evaluation platforms address both by forecasting the genuinely unknown future [Garza et al., 2026, Meyer et al., 2025a], but they can only answer questions about data generated *from now on*: they cannot say how well the models released over the past several years actually forecast the data that has already materialized since their release, and their leaderboards need months to years of operation to accumulate statistical power. We introduce **LiveTS**, which closes this gap with *release-date-stratified retrospective evaluation*: a conservative public release date is registered for every model, and a model is scored only on forecast windows whose evaluation cutoff falls strictly after that release date. Applied to archived history, this rule reconstructs — exactly, for weight-frozen models — the clean scores a live platform would have produced had it been running since each model’s release, for all models at once, today. LiveTS instantiates the protocol on six public real-time domains (energy, weather, air quality, traffic, crypto/FX, and web traffic) collected continuously with point-in-time snapshot semantics and hashed raw snapshots. We pre-registered the full protocol — leakage boundaries, metrics (MASE/CRPS/WQL), geometric-mean aggregation with bootstrap confidence intervals, and Diebold–Mariano tests with Holm correction — before running the main experiments. Using historical simulation over 205 daily series and three rolling cutoffs (50,284 forecast windows), we report the first *clean* scores for six open TSFMs against statistical baselines and supervised baselines retrained strictly before each cutoff. We find no universal winner: TSFMs deliver consistent gains on seasonal domains but fail to beat a seasonal-naïve baseline on financial series, and rankings shift across cutoffs — evidence that single-number leaderboards on static benchmarks overstate what zero-shot forecasting has achieved. LiveTS ships with an open collection pipeline, audit-ready snapshot hashes, and a monthly-refresh live track that hands over to genuine future-only evaluation, complementing — rather than competing with — existing live platforms: retrospective stratification answers the historical question immediately, and the live track keeps the answer honest going forward.

1 Introduction

Benchmark evaluation in time series forecasting is facing a quiet crisis. The field’s standard test sets — ETT, Electricity, Traffic, Weather, the Monash archive [Godahewa et al., 2021] — were assembled years before the current generation of time series foundation models (TSFMs) was trained [Ansari et al., 2024, Das et al., 2024, Woo et al., 2024, Shi et al., 2025]. Modern TSFMs are pretrained on

web-scale corpora of public time series; the very datasets used to certify their “zero-shot” ability are, at least in part, plausibly inside their training distribution, and in the worst case inside their training set verbatim [Meyer et al., 2025b]. Reported zero-shot numbers are therefore upper bounds of unknown looseness.

A second, older problem compounds the first. Static test sets are evaluated millions of times by a community that collectively tunes architectures, hyperparameters, and random seeds against them. Even with honest individual practice, the community-level process is adaptive data reuse, and the resulting benchmark gains need not transfer.

Both problems share a root cause: *the test data existed before the models did*. The community has recently recognized that the fix must be structural, not statistical: live evaluation platforms score forecasts of the genuinely unknown future, making contamination impossible by design. Impermanent [Garza et al., 2026] evaluates models sequentially on continuously updated GitHub activity streams; TS-Arena [Meyer et al., 2025a] enforces a forecast pre-registration protocol on live energy data, requiring predictions to be committed before the ground truth physically exists.

Live platforms, however, answer only a prospective question. By construction they evaluate data generated *after the platform started*: they cannot say how well the TSFMs released over 2024–2025 actually forecast the two years of data that have already materialized since those releases, and a new platform needs months to years of operation before its leaderboard accumulates enough windows for statistically meaningful comparisons. Meanwhile each platform is anchored to one domain (GitHub activity; energy), so live scores may reflect domain idiosyncrasies as much as general forecasting ability.

LiveTS closes this gap with *release-date-stratified retrospective evaluation*. Every model registers a conservative, publicly verifiable release date for its weights; a model is scored only on forecast windows whose evaluation cutoff falls strictly after that date. For weight-frozen models this replays, against archived multi-domain history, exactly the evaluation a live platform would have produced had it been running since each model’s release — for all historical models at once, with two years of ground truth already in hand. Contamination remains impossible for the same structural reason (the model cannot have been pretrained on data that did not exist when its weights were frozen), and community overfitting is ruled out because the clean windows postdate every architecture and hyperparameter decision embodied in the released weights.

Our contributions:

1. **Release-date-stratified retrospective evaluation** (§3): registered model release dates, rolling cutoffs, point-in-time (PIT) collection semantics, and hashed raw snapshots for audit — a protocol that delivers live-grade leakage guarantees on already-materialized data, immediately, for every historical model.
2. **A pre-registered protocol** (§3.3, full text in Appendix A): metrics, aggregation, uncertainty quantification, significance testing, and an amendment policy, frozen before the main experiments.
3. **First stratified clean scores** (§4): 205 daily series from six domains (versus single-domain live platforms), three rolling cutoffs, six open TSFMs, statistical baselines (seasonal naive, AutoETS), and supervised baselines (DLinear, PatchTST, iTransformer) retrained strictly before each cutoff — 50,284 forecast windows with full JSONL provenance.
4. **A live track that hands over to the future** (§5): an open-source collection pipeline running daily across redundant free sources and a monthly pre-registered round whose ground truth is produced by reality after submissions close — designed to interoperate with, not replace, existing live platforms.

2 Related Work

Static TSFM benchmarks. GIFT-Eval [Aksu et al., 2024] and fev-bench [Shchur et al., 2025] aggregate large collections of public datasets and standardize protocol details, but both are static: their test sets predate the models under evaluation, so pretraining overlap must be handled by best-effort de-duplication, which cannot rule out paraphrased or derived copies [Meyer et al., 2025b]. The

Monash archive [Godaheewa et al., 2021] is the canonical pre-TSFM benchmark and is now saturated. LiveTS is complementary: it trades corpus breadth for a guarantee no static benchmark can offer.

Live time series evaluation. Impermanent [Garza et al., 2026] scores eleven models (statistical baselines and current TSFMs) sequentially on daily-updated GitHub activity series with a rolling public leaderboard; TS-Arena [Meyer et al., 2025a] introduces a forecast pre-registration protocol — predictions must be committed before the ground truth exists — operated for a year on energy series. Both make contamination impossible *prospectively*. LiveTS differs on three axes. (i) *Direction of time*: release-date stratification evaluates the models of the past on the (archived) future of their release dates, answering today a question live platforms can only begin answering months from now; our historical simulation already spans up to three clean semi-annual cutoffs per model. (ii) *Domain breadth*: six domains versus one (GitHub activity; energy), separating general forecasting ability from domain idiosyncrasy (§4 shows rankings genuinely differ across domains). (iii) *Interoperability*: our live track (§5) applies the same pre-registration logic going forward, so retrospective and live scores share one protocol; where model sets overlap, comparing a model’s LiveTS retrospective stratum with its live-platform score gives an external consistency check we intend to report as both leaderboards accumulate overlapping models. The M6 competition [Makridakis et al., 2025] ran a genuine live forecasting evaluation but was a one-off event in a single (financial) domain; LiveBench [White et al., 2025] applies the future-only insight to language models.

Evaluation methodology. A line of work documents protocol inconsistencies (look-back lengths, normalization, per-dataset tuning) that make cross-paper comparisons unreliable [Meyer et al., 2025b]. Meyer et al. [Meyer et al., 2025b] analyze the train–test lineage of prominent TSFMs and document both direct sample overlap and indirect temporal overlap (correlated series from the same period leaking future information); they call for evaluation on truly out-of-sample future data. LiveTS is one concrete answer: it fixes a single pre-registered protocol, forbids per-dataset tuning, and combines PIT data semantics with registered release dates. Release-date stratification also addresses the temporal-overlap channel: for windows after a model’s release, no series — correlated or not — from the evaluation period can have been in its training data.

Contamination auditing. Membership-inference and overlap-detection methods estimate contamination post hoc. LiveTS gives the complementary counterfactual: the gap between a model’s score on old benchmarks and its clean LiveTS score is an upper bound on what contamination plus community overfitting bought (we develop this analysis jointly with a companion contamination-audit project).

3 LiveTS Design

3.1 Data layer

Six domains, chosen for public availability, daily-or-finer update frequency, key-free access, and at least two redundant sources each: electric grid load/price (NESO UK, SMARD DE), weather station aggregates (Open-Meteo), PM2.5 air quality (Open-Meteo AQ), public transit ridership (NYC MTA/Socrata), crypto and FX rates (Coinbase, ECB/Frankfurter), and Wikipedia pageviews (Wikimedia). A configurable collector runs daily via cron on two geographically separate hosts, stores the raw API response verbatim, tags every record with `collected_at` (PIT semantics), and archives snapshots to object storage with a SHA-256 manifest so that any later revision of source data is detectable. The evaluation set currently spans **205 daily series** (weather 80, web traffic 58, crypto/FX 34, air quality 20, traffic 7, energy 6); the series list is configuration, not code, and grows without changing the protocol.

3.2 Leakage boundary: release dates and future-only scoring

Every evaluated model registers a **release date**: the earliest publicly verifiable timestamp of its weights (we use the Hugging Face repository creation date; when in doubt, the earlier date is used — conservative in the direction of *fewer* clean windows). Given rolling cutoffs $c_1 < c_2 < \dots$, a model’s clean score aggregates only forecast windows with cutoff $>$ release date. Inputs at each forecast origin are strictly pre-origin observations; targets are the subsequent horizon. In live rounds the target values physically do not exist at submission time; the historical-simulation study in §4 replays the

same rule against archived history, which is exact provided the model’s weights were frozen at its release date.

3.3 Pre-registered protocol

Frozen 2026-08-09 (summary; full text Appendix A). Three cutoffs (2025-01-01, 2025-07-01, 2026-01-01); four forecast origins per cutoff drawn from the 180 days following it; horizon 14 at daily frequency; look-back = each model’s maximum context up to 2048 points, identical across domains; no per-dataset tuning of any kind. Metrics: MASE [Hyndman and Koehler, 2006] (seasonal in-sample naive scale; period 7 for seasonal domains, 1 for financial), CRPS approximated from nine deciles [Gneiting and Raftery, 2007], and weighted quantile loss (WQL). Aggregation: arithmetic mean over origins $\times$ seeds within a series, then geometric mean across series; 95% CIs from a cross-series bootstrap ($B=1000$). Sampling-based models run seeds $\{0, 1, 2\}$; deterministic models are flagged. Significance: paired bootstrap and Diebold–Mariano [Diebold and Mariano, 1995] with the Harvey–Leybourne–Newbold small-sample correction [Harvey et al., 1997] on shared windows, Holm-corrected [Holm, 1979] at $\alpha=0.05$. Point-only models receive no probabilistic score (we do not impute quantiles from point forecasts). Supervised baselines are retrained from scratch on strictly pre-cutoff data for every cutoff, making every cutoff clean for them by construction.

3.4 Service layer

Snapshots and manifests live in object storage (Cloudflare R2); a monthly evaluation round opens after each cutoff, accepts one submission per model per round, and publishes scores when the target window has fully materialized. New models must register a release date before their first round; protocol changes are timestamped amendments that never retroactively alter published scores.

4 Historical-Simulation Experiments

4.1 Setup

205 series $\times$ 3 cutoffs $\times$ 4 origins $\times$ horizon 14 ($\approx 2,460$ windows per deterministic model; $\times 3$ seeds for sampling models and supervised baselines). Models: seasonal naive; AutoETS; DLinear [Zeng et al., 2023], PatchTST [Nie et al., 2023], iTransformer [Liu et al., 2024] (global models, retrained pre-cutoff, multi-quantile loss, 3 seeds); Chronos-Bolt-small/base and Chronos-T5-small [Ansari et al., 2024]; Time-MoE-50M [Shi et al., 2025] (point-only); Moirai-1.1-R-small [Woo et al., 2024]; TimesFM-2.5-200M [Das et al., 2024]. All zero-shot inference used a single configuration per model across all domains. Every result row carries the git commit, environment versions, and run timestamp (JSONL, open).

4.2 Main results

Table 1 and Figure 1 report the main results; Table 2 and Figure 2 break them down by domain. Three observations:

1. **Release-date gating matters.** TimesFM-2.5’s clean score covers only the cutoff after its 2025-09 release; comparing it to models with three clean cutoffs on the pooled table would be flattering nonsense. LiveTS makes this visible instead of averaging it away. On shared windows, TimesFM-2.5 is significantly better than every other model after Holm correction except Chronos-Bolt-base, where the difference does not survive correction (Appendix C).
2. **No universal winner.** On financial series (crypto/FX), no TSFM with three clean cutoffs beats seasonal naive on MASE (TimesFM-2.5 does on its single clean cutoff) — consistent with near-martingale dynamics — while on seasonal domains (weather, air quality, traffic, web) TSFMs deliver 10–40% MASE improvements (Figure 2).
3. **Significance survives correction only for the largest gaps.** After Holm correction, Chronos-Bolt (small/base) and Moirai are significantly better than seasonal naive, but Chronos-T5-small and Time-MoE are not; PatchTST is statistically indistinguishable from seasonal naive and AutoETS on shared windows; several pairwise TSFM comparisons that look decisive as point estimates are not significant (Appendix C).

Table 1: Clean scores on LiveTS (historical simulation, 205 series, future-only cutoffs). Geometric mean across series of per-series mean MASE/WQL; 95% cross-series bootstrap CI ($B=1000$). [†] fewer than three clean cutoffs due to release-date gating — compare on overlapping cutoffs only. Supervised baselines (release “per-cutoff”) are retrained strictly pre-cutoff, so every cutoff is clean for them.

Model	Release	geo-MASE [95% CI]	geo-WQL	Series	Windows
timesfm-2.5-200m [†]	2025-09-02	0.789 [0.711, 0.873]	0.120	205	820
chronos-bolt-small	2024-11-25	0.893 [0.817, 0.976]	0.127	207	2484
chronos-bolt-base	2024-11-25	0.895 [0.826, 0.980]	0.127	207	2484
chronos-t5-small	2024-02-21	0.956 [0.877, 1.048]	0.132	205	7380
moirai-1.1-r-small	2024-06-14	0.965 [0.880, 1.051]	0.140	205	7380
time-moe-50m	2024-09-21	1.004 [0.923, 1.111]	—	205	2460
patchtst	per-cutoff	1.099 [1.019, 1.185]	0.179	207	7452
autoets	—	1.133 [1.050, 1.223]	0.212	205	2460
seasonal_naive	—	1.181 [1.092, 1.268]	0.200	205	2460
dlinear	per-cutoff	1.597 [1.428, 1.788]	0.334	207	7452
itransformer	per-cutoff	4.220 [3.860, 4.627]	0.933	207	7452

Table 2: Per-domain geo-MASE. No model wins everywhere; on crypto/FX no model beats seasonal naive except TimesFM-2.5 on its single clean cutoff.

Model	air_quality	crypto_fx	energy	traffic	weather	web_traffic
timesfm-2.5-200m	0.75	2.16	0.93	0.75	0.68	0.54
chronos-bolt-small	0.79	2.66	0.90	1.24	0.70	0.67
chronos-bolt-base	0.79	2.75	0.89	1.19	0.70	0.67
chronos-t5-small	0.83	2.98	0.97	1.46	0.75	0.68
moirai-1.1-r-small	0.83	2.75	1.03	1.61	0.76	0.72
time-moe-50m	0.81	3.60	0.89	1.04	0.80	0.71
patchtst	0.85	2.98	1.02	1.28	0.95	0.82
autoets	1.17	2.57	1.25	1.39	0.93	0.88
seasonal_naive	1.04	2.57	1.30	1.71	1.05	0.87
dlinear	1.02	8.01	1.17	1.46	1.25	1.08
itransformer	2.74	13.61	3.16	9.00	3.46	3.08

Supervised baselines were retrained pre-cutoff with a fixed generic configuration (multi-quantile loss, input size 96, 1000 steps, no per-dataset tuning). PatchTST is competitive with weaker TSFMs; DLinear and iTransformer underperform under this no-tuning regime. iTransformer runs in univariate mode ($n_{\text{series}}=1$), a conservative configuration at odds with its multivariate design; we report it as a lower bound and note this as a limitation rather than tuning it post hoc.

4.3 Robustness

Figure 3 shows per-cutoff scores: mid-table rankings (Chronos-T5 vs. Moirai vs. Time-MoE vs. PatchTST) reorder across cutoffs, while the top (Chronos-Bolt) and bottom are stable. Seed variance for sampling models is small relative to cross-series dispersion. This is quantitative evidence that a single static split materially misranks models — and an argument for rolling evaluation as the default.

4.4 Clean scores vs. post-hoc contamination signals

A companion contamination audit¹ probes the same model families post hoc: for each TSFM it contrasts forecasting skill on *member* datasets (documented in the model’s pretraining corpus) against *non-member* datasets. Table 3 places its skill-probe results next to our clean LiveTS scores.

The audit finds member-dataset skill consistently higher for the Chronos family and TimesFM (Cliff’s δ 0.47–0.85), directionally consistent with contamination-inflated legacy scores, but no effect survives Holm correction at the audit’s sample size — post-hoc detection is statistically underpowered even when the direction is consistent. This is precisely the argument for LiveTS: rather than trying to detect contamination after the fact, future-only evaluation makes it impossible. The two approaches

¹<https://github.com/wookat/tsfm-contamination-audit>; 8 TSFMs $\times$ 14 datasets, member vs. non-member skill probes with permutation tests and Holm correction.

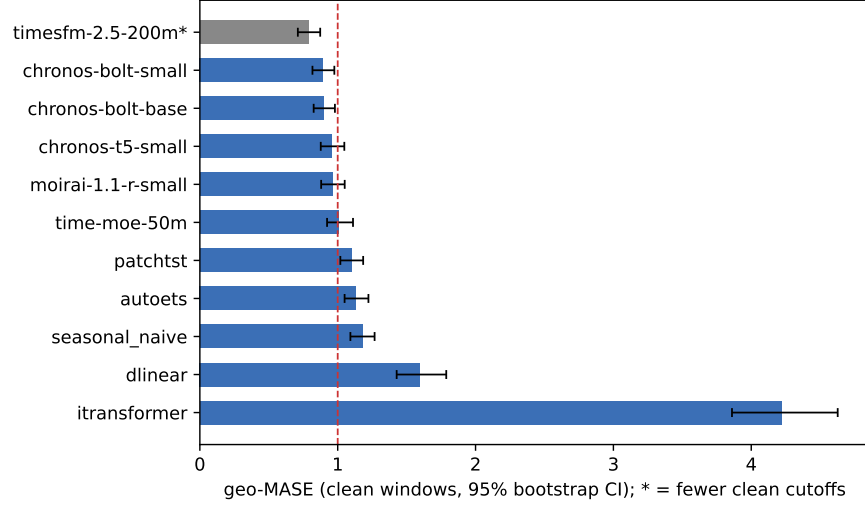


Figure 1: Clean geo-MASE with 95% bootstrap CIs. Grey bar: model with fewer clean cutoffs (release-date gating); dashed line: seasonal-naive parity.

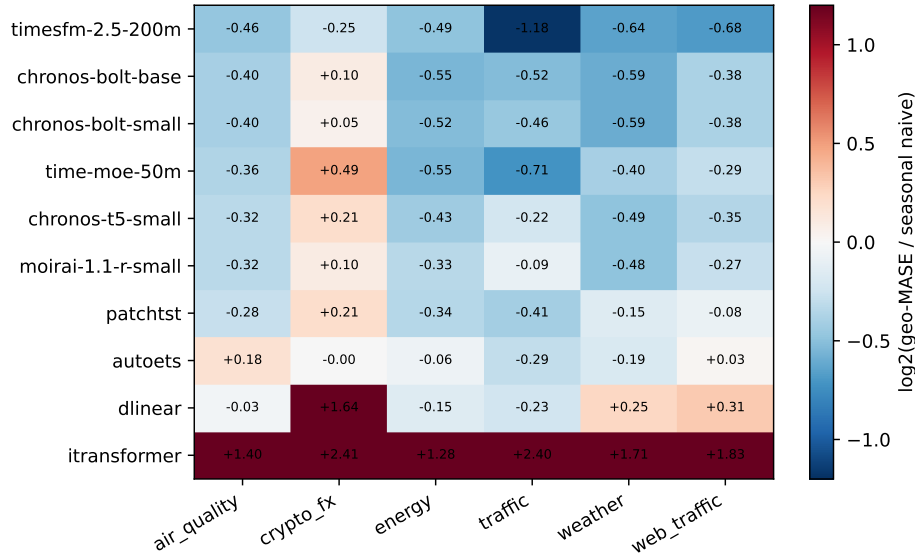


Figure 2: Per-domain geo-MASE relative to seasonal naive ($\log_2$ ratio; blue = better than naive). TSFMs win on seasonal domains and lose on crypto/FX.

are complementary; LiveTS clean scores are currently the only scores for these models that the contamination question cannot touch.

5 The Live Benchmark

The retrospective study answers the historical question; the live track keeps future scores equally clean, applying the same pre-registration logic as TS-Arena [Meyer et al., 2025a] but across all six domains. Round-0 opens with the protocol frozen in Appendix A. Monthly cadence: cutoff on the 1st, submissions within 7 days (forecasts only — no post-cutoff data access), scoring after the 14-day target window completes, publication with full JSONL provenance. Anti-gaming: one submission per

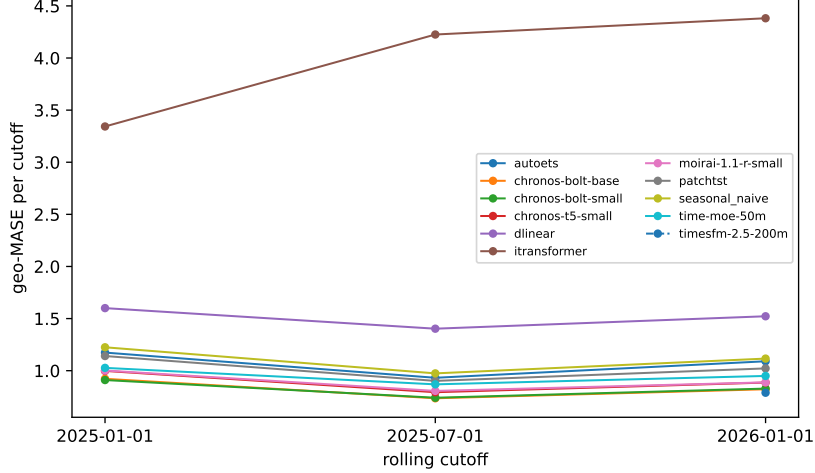


Figure 3: Per-cutoff geo-MASE. Mid-table rankings shift across cutoffs; a single static split would misrank models.

Table 3: LiveTS clean skill vs. post-hoc audit signals. $\text{skill}_{\text{live}} = \text{geo-MASE}(\text{model})/\text{geo-MASE}(\text{seasonal naive})$ on shared clean windows (lower is better). Audit columns: mean skill on member vs. non-member datasets and Holm-corrected permutation p ; a positive member–non-member gap is directionally consistent with contamination-inflated legacy scores. ^aaudited checkpoint differs from the one we evaluate (Moirai-1.0-R-small; TimesFM-2.0-500M) — directional reference only.

Model	$\text{skill}_{\text{live}}$	Audit member	Audit non-member	Holm p
Chronos-Bolt-small	0.754	0.297	−0.673	0.86
Chronos-Bolt-base	0.756	0.323	−1.088	0.62
Chronos-T5-small	0.809	0.253	−0.108	1.00
Time-MoE-50M	0.850	−0.382	0.173	1.00
Moirai-1.1-R-small	0.817	−0.216 ^a	0.026 ^a	1.00
TimesFM-2.5-200M	0.707	0.417 ^a	−0.134 ^a	0.24

model per round; release-date registration precedes first participation; raw snapshot hashes published for third-party audit; the aggregation code is deterministic and open.

6 Limitations and Ethics

Univariate daily-frequency focus (intraday and multivariate tracks are future work); free-tier sources can rate-limit or discontinue (mitigated by ≥ 2 sources per domain, but a domain could still degrade); public infrastructure data skews toward well-instrumented regions; release dates rely on public timestamps and may be later than private availability (our conservative rule errs toward fewer clean windows, not more). Supervised baselines use one generic configuration — their scores are lower bounds, not the best achievable. The benchmark evaluates forecast quality only; downstream decision costs are out of scope. Retrospective stratification is exact only for weight-frozen models: it cannot cleanly score continuously retrained or API-only models whose effective release date is ambiguous — for those, prospective live platforms [Garza et al., 2026, Meyer et al., 2025a] are the right tool, which is why we view the two designs as complementary. Finally, an external consistency check against live-platform leaderboards requires overlapping model sets and accumulation time on both sides; we report point-estimate comparisons as they become available rather than claiming them now.

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A Pre-registered protocol v1.0 + amendment A1

The full pre-registration (frozen 2026-08-09) is distributed with the benchmark repository (docs/protocol-prereg.md); amendment A1 (series expansion 25→205 and supervised baselines) was registered before the expanded runs and does not alter previously published results.

B Data sources and series list

See docs/data-sources.md and the generated series inventory in the repository. Every series has ≥ 800 daily points (≥ 700 for FX, trading-day calendar).

C Full significance matrices

The full pairwise matrix (56 pairs, paired bootstrap $B=10,000$ and DM-HLN, Holm-corrected) is generated by scripts/significance.py and distributed as docs/significance.md.

D Reproduction

One command per table; JSONL schema; environment lockfiles; commit hashes. See REPRODUCIBILITY.md in the repository.